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Waveform inversion of broad-band body wave data for the *S*-velocity structure in the lowermost mantle beneath the Indian subcontinent and Tibetan Plateau

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SUMMARY

We conduct waveform inversion for the radial shear velocity structure of the lowermost mantle beneath the Indian subcontinent and Tibetan Plateau using transverse component body wave waveform data obtained from the ORFEUS (Observatories and Research Facilities for European Seismology) and IRIS (Incorporated Research Institutions for Seismology) arrays in the passband 8–200 s. As these are higher frequency data than used by most previous waveform inversion studies, several improvements in terms of techniques were necessary. We conduct tests to confirm the robustness of the results. The average *S* velocity of this region is almost the same as Preliminary Reference Earth Model (PREM), but we find that the velocity is faster than PREM in the depth range 2400–2750 km and slower than PREM in the depth range 2750 km—core–mantle boundary (CMB). This suggests that the perovskite to post-perovskite phase transition occurs in the depth range 2400–2600 km. If we assume the composition of D″ is pyrolitic, the thickness of the thermal boundary layer at the CMB is about 200–300 km.

Key words: Composition of the mantle; Body waves; Seismic tomography.

GJI Seismology

1 INTRODUCTION

Waveform inversion, sometimes called full waveform inversion, is a powerful technique for extracting all of the information contained in observed seismic body wave waveforms without using ray-theoretical or other asymptotic approximations (e.g. Kawai *et al.* 2007a; Konishi *et al.* 2009; Fichtner *et al.* 2010). To conduct waveform inversion, it is necessary to be able to efficiently and accurately calculate synthetic seismograms and their partial derivatives. Our group has developed the direct solution method (DSM, e.g. Geller & Ohminato 1994; Geller & Takeuchi 1995; Kawai *et al.* 2006) for computing complete synthetic seismograms in a spherically symmetric transversely isotropic earth model without Earth flattening or other such approximations. We also use the formulation of the partial derivatives with respect to Earth structure of Geller & Hara (1993), which also treats anelasticity in a straightforward manner (Fuji *et al.* 2010). Note that we invert only for elastic structure in this study. Using these accurate computational methods, we have conducted inversion of body wave waveform data (e.g. Kawai *et al.* 2007a,b, 2009, 2010; Konishi *et al.* 2009; Kawai & Geller 2010a,b,c; Fuji *et al.* 2010). This objective and automated method, which simultaneously uses a large number of waveforms, allows us to analyse overlapping phases which cannot be easily handled by visual fitting techniques. As compared to other methodologies, waveform inversion does not use model parametrizations with sharp discontinuities. Rather it obtains models without any assumptions about discontinuities and thus obtains robust but relatively smooth models (see the discussions in Konishi *et al.* 2009).

Most of the above waveform inversion studies were conducted for the 20–200 s passband. As this passband is relatively long period, knowledge of the details of the source time functions was unnecessary, and they were approximated as δ-functions. However, to maximize the resolving power, it is preferable that we use as broad a passband as possible in waveform inversion. Kawai & Geller (2010a) introduced 'direct inversion method' to use the 8–200 s passband in waveform inversion. Because the δ-function source time function approximation is not appropriate for this passband, they used boxcar moment rate functions whose half duration was obtained from the global CMT solutions. In this study, we also consider the 8–200 s passband in waveform inversion. However, as our data set contains stations at epicentral distances near the shadow zone, we must carefully consider the static correction methods, as they can be susceptible to non-linear effects.

1.1 D″ region

The region at the bottom of the lowermost mantle is called D″ (Bullen 1949). A seismic discontinuity at the top of D″ was found

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by Lay & Helmberger (1983). This discontinuity is now known to be due to the phase transition from perovskite (pv) to post-perovskite (ppv; Murakami *et al.* 2004). Because D″ plays a crucial role as a thermal boundary layer in the deep mantle, it is important to precisely determine regional models of D″ for as many regions as possible. The first applications of body wave waveform inversion (Kawai *et al.* 2007a,b) were to study D″ regions which have an average *S* velocity higher than Preliminary Reference Earth Model (PREM; Dziewonski & Anderson 1981). Konishi *et al.* (2009) and Kawai & Geller (2010a) then studied D″ in regions beneath the Pacific which have average velocity lower than PREM. We have not yet, however, analysed D″ for a region with an average velocity close to PREM (see, e.g. Garnero, 2000; Takeuchi 2007). That is the purpose of this study.

Many previous studies used methods other than waveform inversion to investigate the lowermost mantle beneath Eurasia to search for a deep mantle discontinuity (Lay & Helmberger 1983; Weber & Davis 1990; Gaherty & Lay 1992; Houard & Nataf 1992, 1993; Wysession *et al.* 1992; Weber 1993; Scherbaum *et al.* 1997; Thomas & Weber 1997; Bilek & Lay 1998; Thomas *et al.* 2004). The lowermost mantle beneath the Indian Ocean has also been investigated using methods other than waveform inversion (Young & Lay 1987; Wen 2001; Wright & Kuo 2007). Some studies (e.g. Thomas *et al.* 2004) reported evidence for two sharp velocity gradients in D″ beneath northern Eurasia. The double crossing hypothesis (Hernlund *et al.* 2005) could explain such structure. However, Kawai & Tsuchiya (2009) showed that structure at the base of the mantle can be explained by thermal effects such as a steep negative gradient of shear velocity due to a thermal boundary layer, thus obviating the need for invoking a double crossing.

In this study, we conduct waveform inversion for shear velocity structure in the lowermost mantle beneath India and north of southeastern Asia and western China. Young & Lay (1987) and Gaherty & Lay (1992) studied the northern and southern parts of this region, respectively. Both of these studies reported the existence of velocity discontinuities in the respective study regions. According to Wysession *et al.* (1992), the *S*-wave velocity in the lowermost 190 km of the mantle beneath Indonesia is 1.0 per cent faster than PREM. In this study, we investigate the vertical velocity structure in these regions without imposing *a priori* velocity jumps. By elucidating the fine structure of this region, which, as noted earlier, has an average D″ velocity close to the global average, we obtain new evidence for further testing the hypothesis of Kawai & Tsuchiya (2009).

2 METHODS

2.1 Waveform inversion

We conduct waveform inversion to infer the seismic structure of the lowermost mantle beneath the Indian subcontinent and Tibetan Plateau. Waveform inversion can deal with observed seismic data objectively and quantitatively. We show the overall flowchart of waveform inversion in Fig. 1. First, we choose a target region and ray paths that sample the target region in this study and download observed waveforms to construct the data set. After selecting high-quality waveform data, we compute synthetic waveforms and

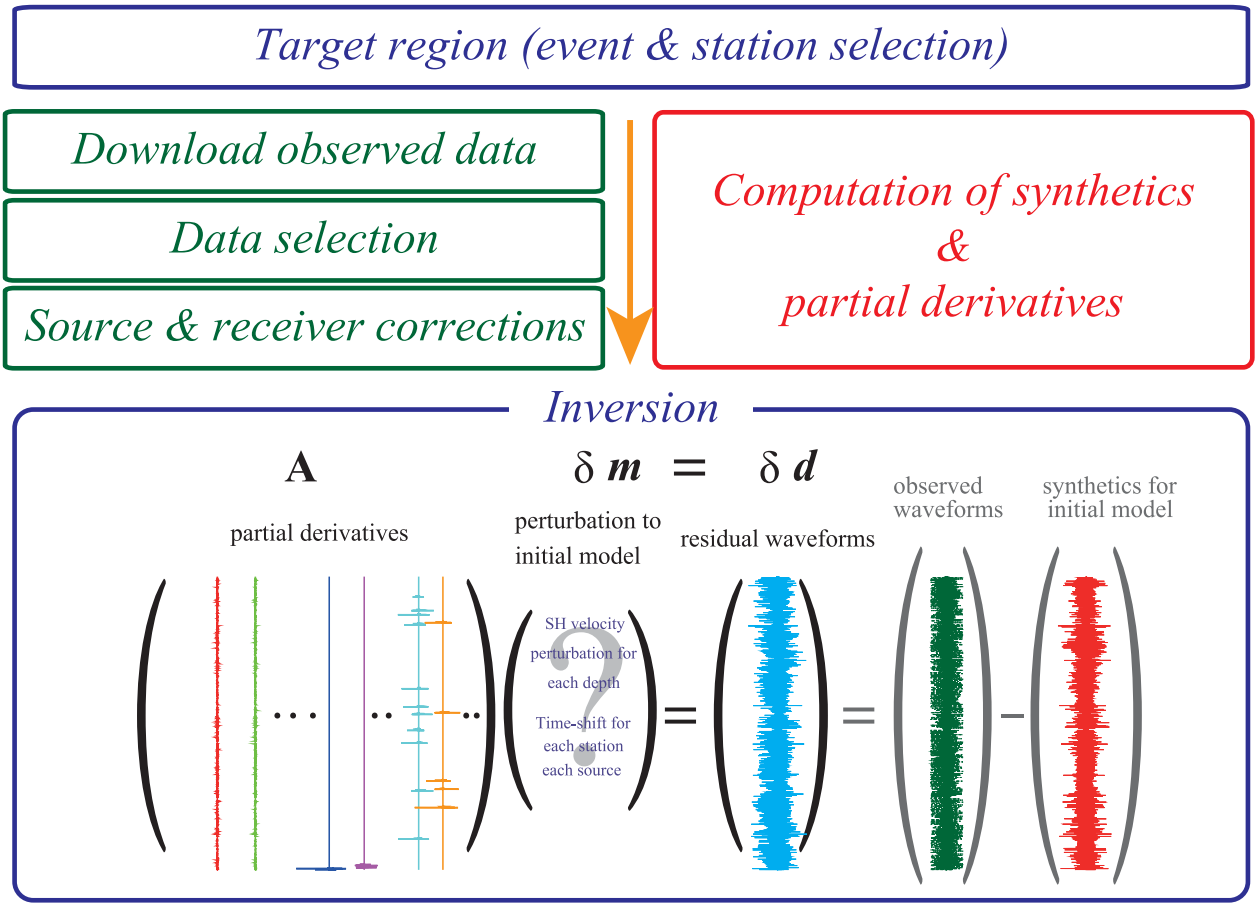


Figure 1. Key steps in waveform inversion. The bottom box schematically depicts the observational equation (eq. 4).

partial derivatives for the initial model. We correct for the effect of local structure near the stations and sources by making static corrections (time-shifts), and check whether each of the observed seismograms satisfies the criteria for inclusion in the data set. Finally, we obtain a new model by inversion using the conjugate gradient (CG) method. The details of each of these steps are discussed later.

2.1.1 Data set and synthetics

We collect 2743 waveforms from IRIS (Incorporated Research Institutions for Seismology; http://www.iris.edu/hq/) and ORFEUS (Observatories and Research Facilities for European Seismology; http://www.orfeus-eu.org/) for sources deeper than 200 km in southeastern Asia which sample the D″ region beneath the Indian subcontinent and Tibetan Plateau. The geometry of the data set is shown in Fig. 2 and information on the events is listed in Table 1. Figs 3 and 4 show the distributions of epicentral distances and turning depths of the paths. We deconvolve the instrumental response, and use transverse component data with a time window that begins 10 s before the *S* arrival and ends 60 s after the *ScS* arrival or the depth phase arrival, whichever is later (using the traveltimes for PREM). The epicentral distances used in this study range from 60° to 95°.

We compute synthetic seismograms (Geller & Ohminato 1994; Kawai *et al.* 2006) and their partial derivatives (Gaherty & Lay 1992) for the initial model, anisotropic PREM (Dziewonski & Anderson 1981) using the source parameters from Global CMT solution, assuming the source time function to be a boxcar moment rate function. We compute the partial derivatives with respect to the *S*-wave velocities at 20 km intervals for the depth range 2391–2891 km (core–mantle boundary, CMB), and fix the model to PREM above a depth of 2391 km. We invert for models using data in the passbands 8–200, 12.5–200 and 20–200 s, to see whether or not there is significant frequency dependence.

Table 1. Earthquakes used in this study.

Event#	Y/M/D	Depth	Lat	Lon	M_w	No. of records used
1	1996 August 22	614.7	−7.14°	123.52°	5.9	3
2	1997 July 11	592.1	−5.70°	110.65°	5.9	5
3	1997 December 2	591.9	6.79°	123.58°	5.5	3
4	1998 May 23	628.8	8.29°	123.86°	6.0	4
5	2000 August 7	648.3	−6.95°	123.53°	6.5	3
6	2000 September 09	222.0	6.50°	95.51°	5.6	9
7	2002 June 16	217.1	−2.26°	102.51°	5.7	13
8	2002 October 3	316.7	−7.59°	115.71°	6.0	5
9	2003 May 26	579.7	6.90°	123.85°	6.8	5
10	2003 July 1	604.6	4.69°	122.67°	6.0	7
11	2004 June 6	600.6	−6.10°	113.15°	5.8	3
12	2005 September 4	454.4	3.05°	123.32°	6.0	7
13	2006 July 15	385.1	−4.39°	126.31°	5.8	3
14	2006 December 1	208.4	3.46°	99.05°	6.3	8
15	2006 December 12	215.3	3.83°	125.10°	6.3	9
16	2007 May 9	569.2	6.62°	123.91°	5.6	5

2.1.2 Data selection

To exclude data with a poor signal-to-noise (S/N) ratio, we compute the ratio of the maximum amplitude of the data to the corresponding synthetic, and eliminate records for which the ratio is greater than 2 or less than 0.5. We also eliminate data for which time-shifts of more than 10 s give the best correlation coefficients or for which the best correlation coefficient is less than 0.5, and events which have no more than 3 observed waveforms satisfying the above criteria, to conduct inversion with a sufficient number of high-quality waveforms for each event. The data set consists of 92 records that satisfy the above criteria. This is a much smaller number of records than our group's previous studies using stations in North America,

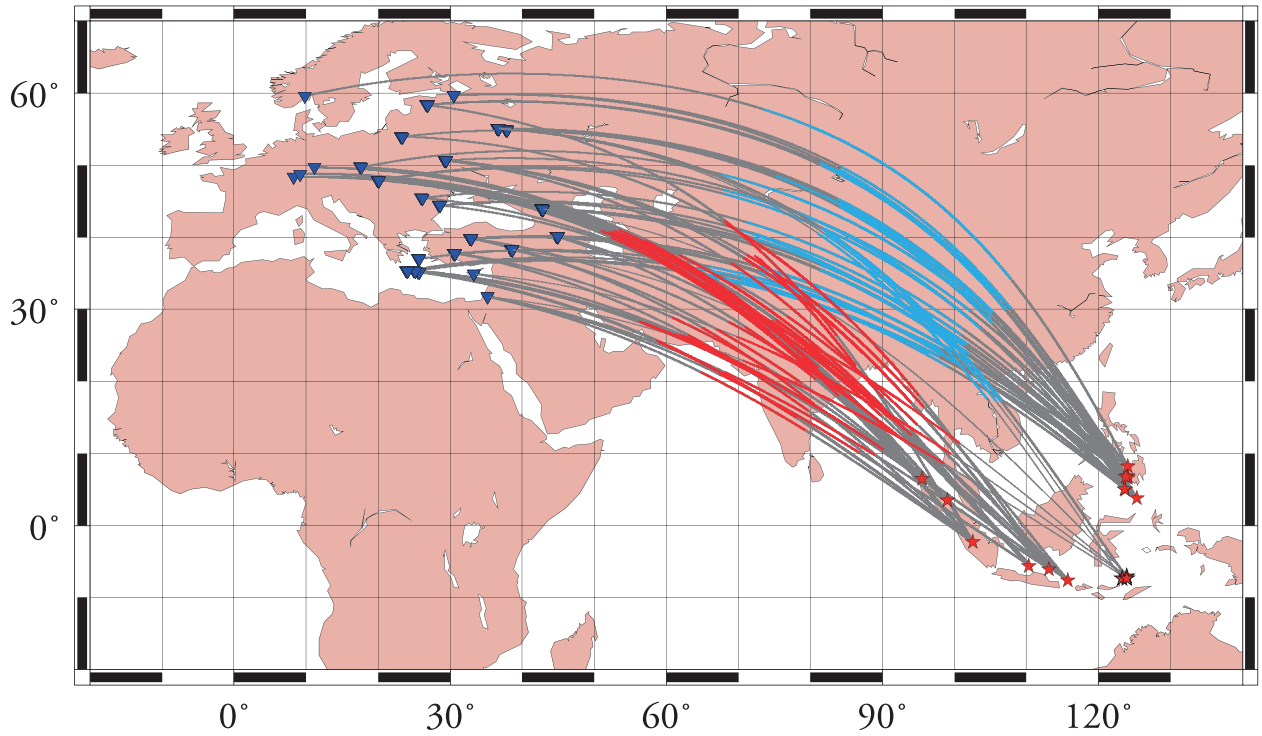


Figure 2. Event-receiver geometry, with great circle ray paths. The portions of the great circles which sample D″ are shown in red and sky blue, respectively, for Regions 1 and 2 (See Fig. 14 and discussion thereof in text). Blue triangles show the stations, and red stars show the earthquakes.

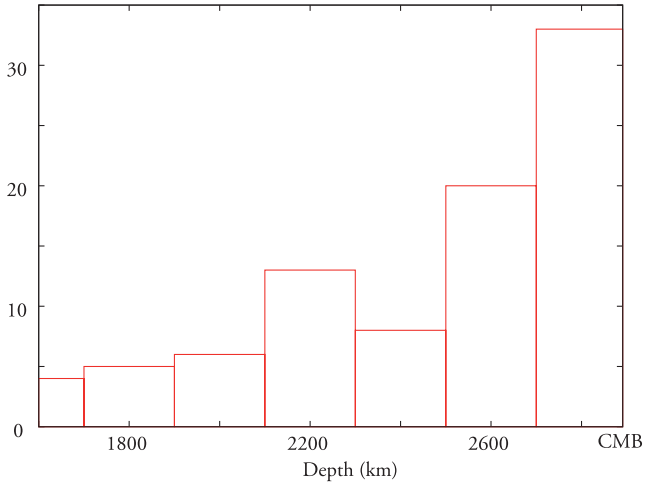


Figure 3. The histogram of *S*-wave turning depth distribution of the data set used in this study.

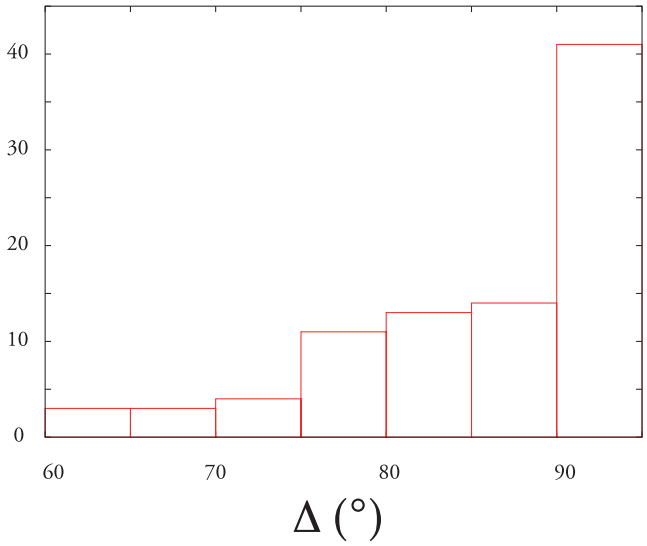


Figure 4. The histogram of epicentral distribution of the data set used in this study.

but is sufficient to obtain a robust model. This is confirmed below by checkerboard studies.

2.1.3 Weighting and scaling method

The reciprocal of the maximum amplitude of each record is used as the weighting factor in the inversion, so that all data have roughly the same importance. The data set is sampled at 1 Hz. To solve the inverse problem, we use the CG method, following Fuji (2010). As we use partial derivatives for not only elastic constants but also for the centroid time, we must consider the relative scaling of the variables. We use units of 1 GPa for $\delta\mu$ and 1 s for δt where μ is around 290 GPa in the D″ region, to roughly normalize the amplitudes of those partial derivatives. See Fuji (2010) for further discussion on the scaling of the model parameters.

2.2 Static correction

In our group's earliest papers on waveform inversion (Kawai *et al.* 2007a,b), we used relatively long-period waveforms (20–200 s) and we approximated the source time function to be a δ-function. Kawai & Geller (2010a) was the first body wave waveform inversion study that attempted to use short-period waveforms with a passband of 8–200 s. To resolve the difficulties concerning the source time functions, we used a boxcar source time function and derived static corrections using the 'direct inversion' method, which simultaneously inverts for the effects of near-source and near-receiver local effects. This was necessary because the 'autopick' method proposed by Fuji *et al.* (2010) cannot be applied directly to waveforms for which the direct *S* phase is also sensitive to the target structure. In this paper, we therefore introduce a new 'hybrid' method for making static corrections. We summarize the direct inversion method, autopick method, and our new hybrid method below. We show that the latter two methods yield robust results.

2.2.1 Direct inversion method

Kawai & Geller (2010a) separated the estimation of the time-shift correction into separate terms for the near-source and near-station effects. They assumed that waveforms from the same source or to the same station have the same static corrections, (δt_e) and (δt_s), respectively, and computed the partial derivatives to include δt_e and δt_s as variables in the inversion. This method can be used when we cannot use the *S* phase as a reference phase, but as our data set has relatively larger epicentral distances, the partial derivatives of the synthetics with respect to time and elastic parameters are very similar. Furthermore, as shorter period data were used in this study, the effects of non-linearity may be more significant.

2.2.2 Autopick method: time-shift before inversion

The 'autopick' method accounts for the effect of structure outside the target region by making a single static correction (time-shift) for each record for each event based on a comparison of the observed and theoretical *S*-wave waveforms. (See fig. 5 of Fuji *et al.* 2010, and their discussion thereof, for details.) In this method, we pick the *S*-wave arrival and first peak for all synthetics and measure the time difference between them, $T_{\pi/4}$, to take the time window from $4T_{\pi/4}$ before the *S*-wave arrival to $T_{\pi/4}$ after the *S*-wave arrival for computing the cross-correlation. The lag time that produces the best cross-correlation for this time window is chosen as the time-shift. After making the static correction, the structure inside the target region can then be studied using later phases that sample the target region, such as *ScS* in the case of D″.

2.2.3 Hybrid method

The autopick method yields satisfactory results when the direct *S*-wave does not sample D″, which is the case for epicentral distances of about $\Delta < 80^\circ$. On the other hand, for epicentral distances of about $\Delta > 80^\circ$, both the direct *S* wave and later phases sample D″. The use of the autopick method for stations at epicentral distances of about $\Delta > 80^\circ$ is thus inappropriate. We therefore have developed the following 'hybrid method' for stations at epicentral distances of $\Delta > 80^\circ$.

The hybrid method proceeds by first applying the autopick method to stations at epicentral distances of $\Delta < 80^\circ$. We obtain a set of station corrections, δt_{ij}, for each record, where i is the index for the event and j is the index for the station. We derive empirical time-shifts for the ith event and for the jth station by averaging over all the stations at epicentral distances $\Delta < 80^\circ$. We then define the average correction for the ith earthquake at epicentral distances

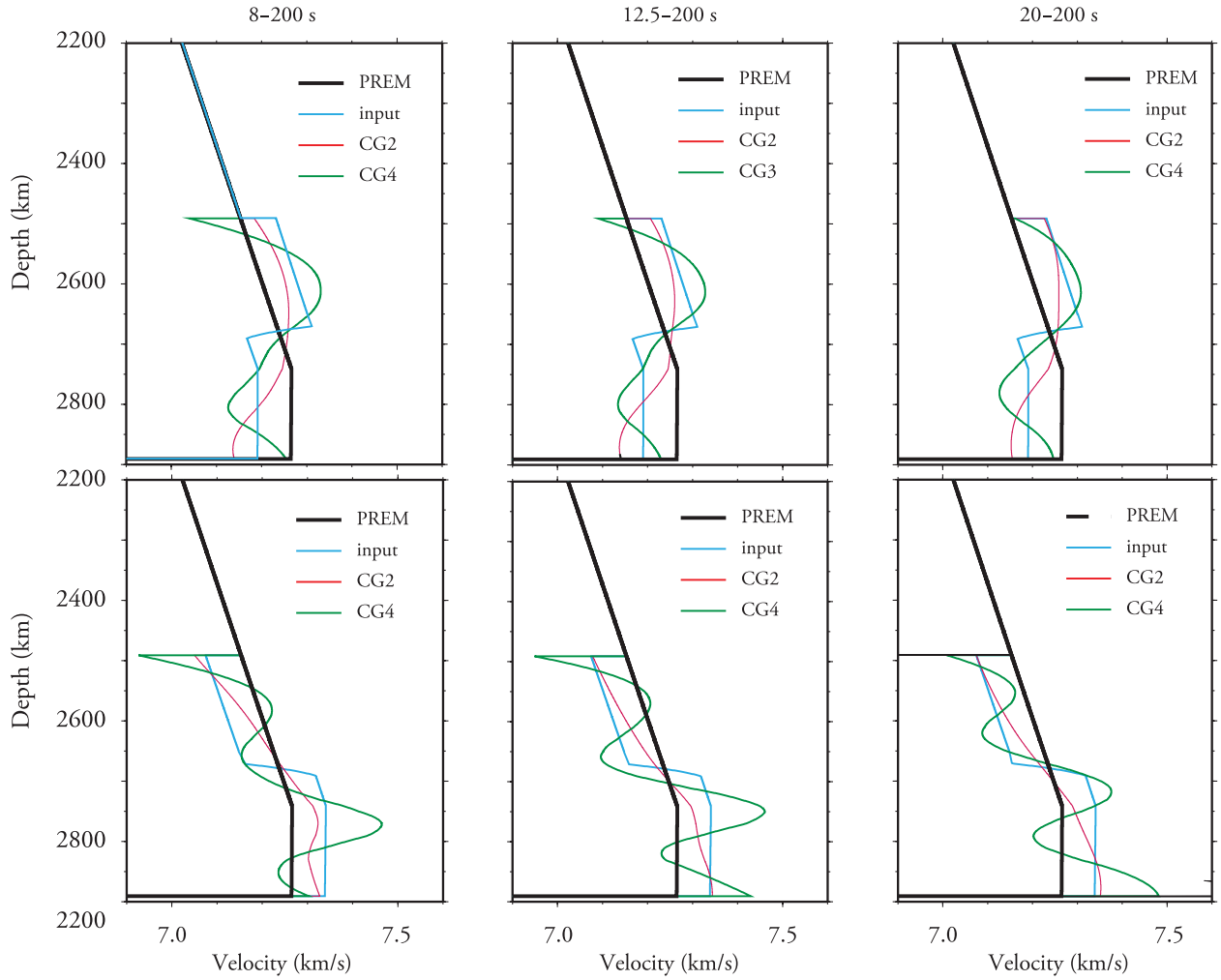


Figure 5. 'Checkerboard' resolution test for our inversions for D″ structure. The blue curves are the models used to compute the synthetic 'observed data'. The starting model for the inversions (PREM) is shown by the black curves in each panel. The thick green curves are the inversions with respect to PREM for the data set of the passband written above each panel which give the best AICs. The number of the CG vectors for the best AIC is shown in each panel. The models obtained using two CG vectors are also shown as red curves. All of the inversions reasonably recover the starting models.

Table 2. Variance and (AIC) for each static correction method. Each 'PREM' column shows variance and AIC for waveforms with each of the three static correction methods. Columns labelled 'best model' show variance for waveforms for the best models.

	8–200 s		12.5–200 s		20–200 s	
	PREM	Best model	PREM	Best model	PREM	Best model
Autopick	53.6 (1967)	53.2 (1962)	80.0 (1483)	64.3 (1374)	71.4 (888)	59.9 (827)
Direct	178 (3032)	146 (2878)	196 (1995)	155 (1883)	176 (1208)	128 (1117)
Hybrid	178 (3032)	158 (2946)	134 (1619)	104 (1517)	125 (1089)	93.3 (984)

$\Delta < 80^\circ$, δe_i, as follows:

$$\delta e_i = \text{Average}_j(\delta t_{ij}), \quad (1)$$

where the average is taken over all stations for which records at epicentral distances $\Delta < 80^\circ$ are available for at least three earthquakes. We similarly define the average correction for the jth station at epicentral distances $\Delta < 80^\circ$, δe_j, as follows:

$$\delta s_j = \text{Average}_i(\delta t_{ij} - e_i), \quad (2)$$

where the average is taken over all earthquakes for which records at epicentral distances $\Delta < 80^\circ$ are available for at least three stations. We finally define the empirical correction for the ith earthquake at the jth station at epicentral distances $\Delta > 80^\circ$ to be

$$\delta t_{ij} = e_i + s_j. \quad (3)$$

By using the above procedure, we can in principle prevent double counting for source- and receiver-side effects and can therefore expect that data processed in this manner would, in an ideal world, contain purely information on the structure of the deep Earth.

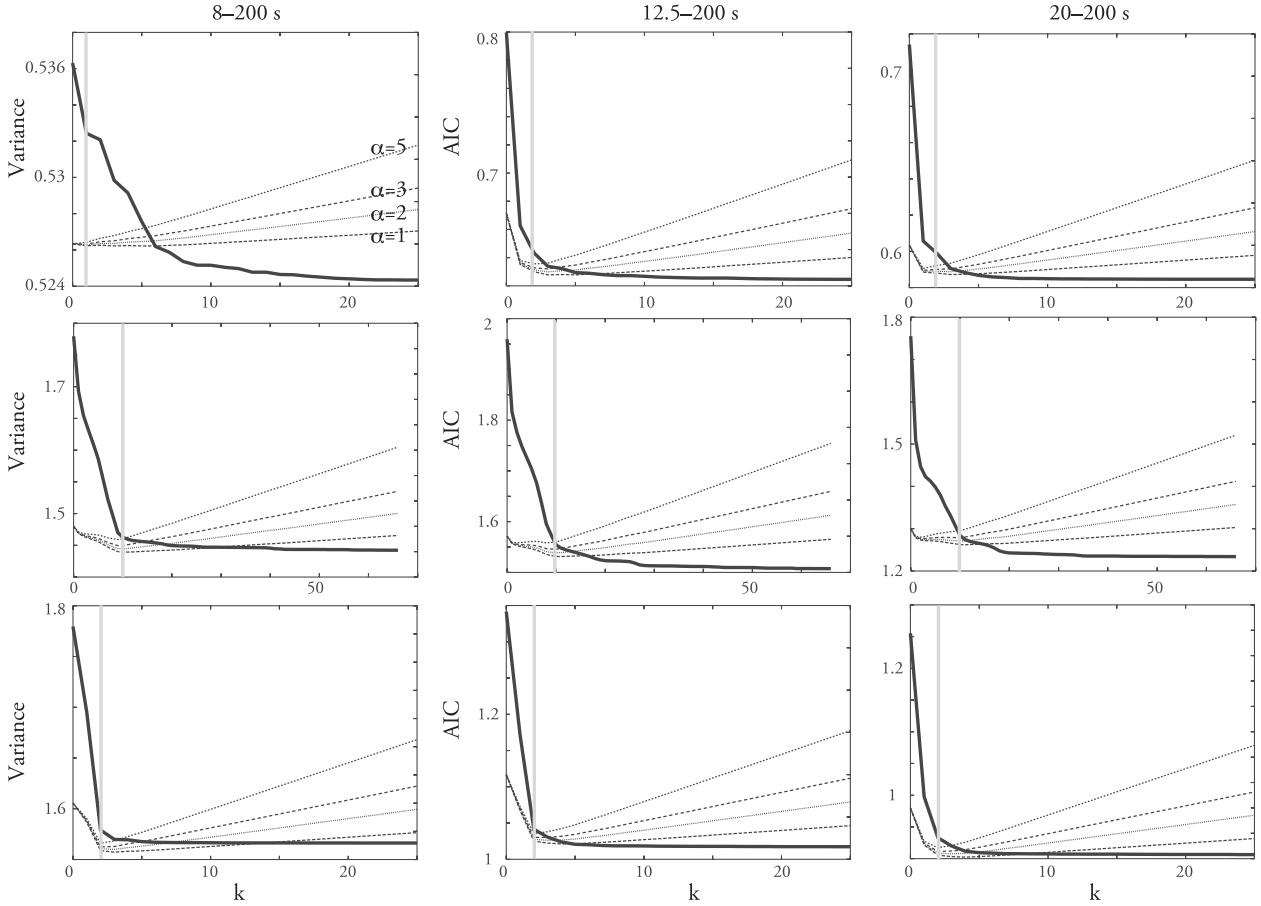


Figure 6. The AIC for each α (thin curves) and variance for each inversion (thick curves). Each column shows the results for the passband written above the top panel. Each row shows the results for one time-shift method (Top: autopick; Centre: direct inversion and Bottom: hybrid). Each panel shows the AIC and variance obtained by using each number k of CG basis vectors. The horizontal axis shows the number of the CG vectors used. Each AIC is calculated for n/α, where n is 12.5 per cent, 8 per cent and 5 per cent of the total number of data points for the respective passbands. The vertical grey lines written in the panels indicate the AIC value.

To confirm this, we consider the above three correction methods and study the robustness of the obtained models.

2.3 CG method

We conduct waveform inversion using the CG method. We write the observational equation (see Fuji *et al.* 2010 and Fuji 2010 for details) as

$$\mathbf{A}\delta\boldsymbol{m} = \delta\boldsymbol{d}, \tag{4}$$

where $\mathbf{A}$ is the matrix of partial derivatives, $\delta\boldsymbol{m}$ is the vector of perturbations to the starting model and $\delta\boldsymbol{d}$ is the vector of residuals. It is well known that this equation is overdetermined and cannot be solved in a rigorous sense, and that a solution is obtained by solving the normal equations, which are written as follows:

$$\mathbf{A}^T\mathbf{A}\,\delta\boldsymbol{m} = \mathbf{A}^T\delta\boldsymbol{d}, \tag{5}$$

we define a sequence of M mutually orthogonal conjugate vectors (CG vectors) $\boldsymbol{p}_i (i = 1, ..M)$,

$$\boldsymbol{p}_j^T\mathbf{A}^T\mathbf{A}\,\boldsymbol{p}_i = 0 \quad \text{for} \quad j \neq i \tag{6}$$

and then we write the solution by summing over the first k vectors

$$\delta\boldsymbol{m} = \sum_{j=1}^{k} a_j\,\boldsymbol{p}_j. \tag{7}$$

We choose the value of k which minimizes the Akaike Information Criterion (AIC; defined below).

2.4 AIC

The AIC (Akaike 1977) is defined as follows:

$$\begin{aligned}\text{AIC}\ (k) &= -2(\text{Maximum Log-Likelihood}) + 2k \\ &= n\log 2\pi + n\log\sigma^2 + n + 2(k+1),\end{aligned} \tag{8}$$

where $k + 1$ is the number of free parameters (the number of unknown coefficients plus one, where one is added because the variance of the residuals is another unknown), n is the number of independent data points, and σ^2 is the variance of the residuals. AIC gives an unbiased estimate of the mean log-likelihood of the model for a given data set. AIC in effect penalizes an increase in the number of model parameters by requiring a commensurate improvement in the variance reduction before a more complex model can be accepted as meaningful.

2.4.1 Number of independent data points for AIC

In this paper, k denotes the number of CG vectors used in the waveform inversion. We assume that n is 12.5 per cent of the data points (at 1 Hz sampling) for the 8–200 s passband data set, 8 per cent

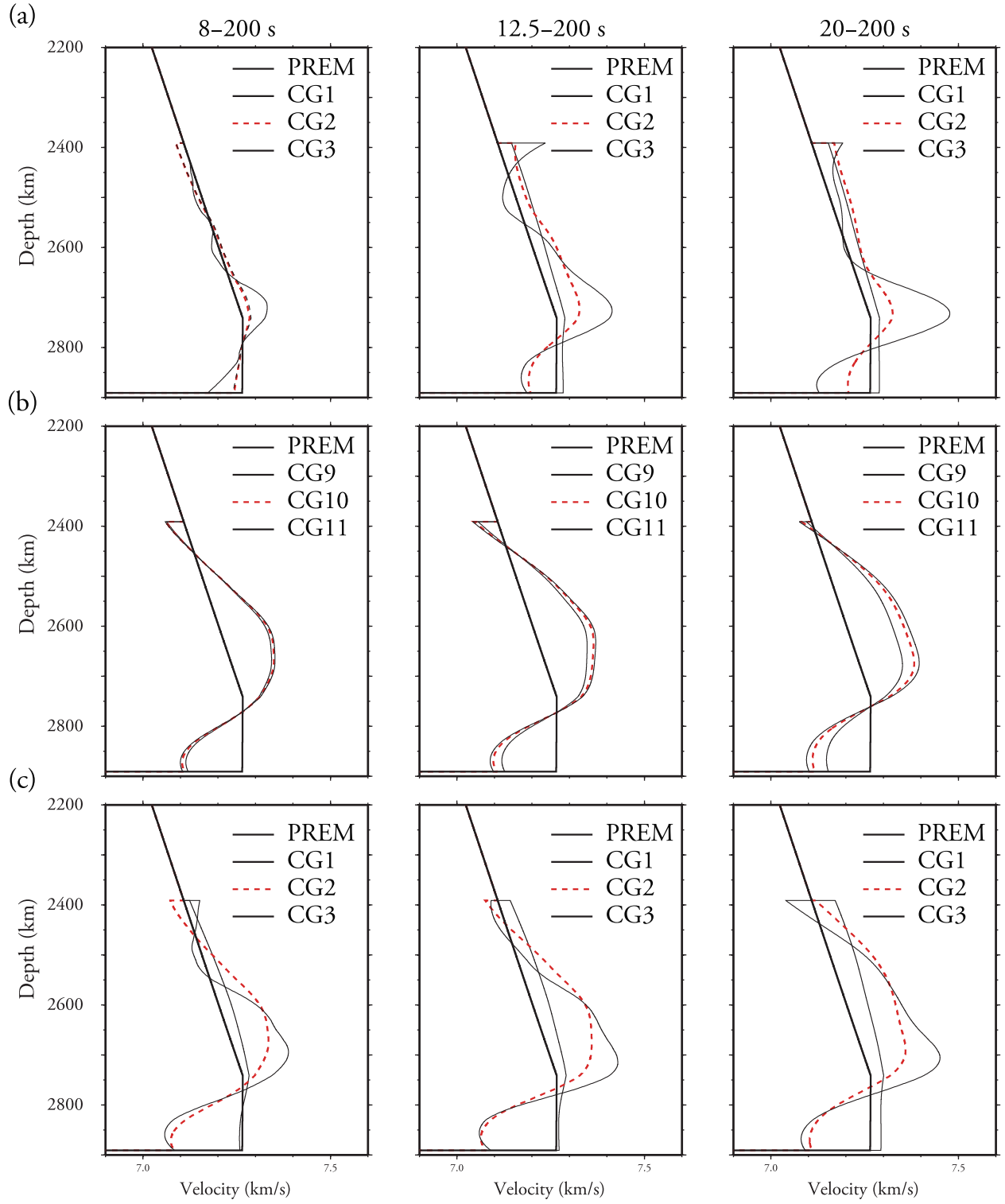


Figure 7. Each column shows the result of the inversions for the passband written above the top panel. The rows show the model for (a) autopick, (b) the direct inversion and (c) the hybrid method. The number of CG vectors used is also shown. The red curves show the best model according to AIC. PREM is also plotted as a black line for reference.

of that for the 12.5–200 s and 5 per cent of that for the 20–200 s passband data set, since the data are filtered in each passband, respectively, thereby reducing the amount of information. However, because the nominally independent data points actually have redundancy between the various waveforms, we follow Fuji *et al.* (2010) by introducing the empirical redundancy parameter, α, defined as

$$n' = \frac{n}{\alpha}, \tag{9}$$

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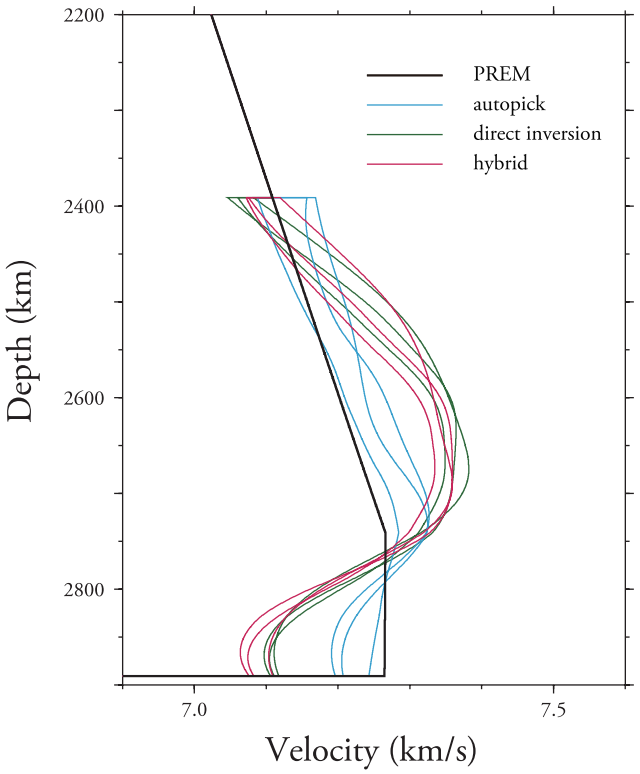


Figure 8. The models with the best AIC for each of the nine panels of Fig. 7 are compared. The blue curves are the models for autopick, the green curves are for direct inversion and the red curves are for the hybrid method. The black line shows PREM.

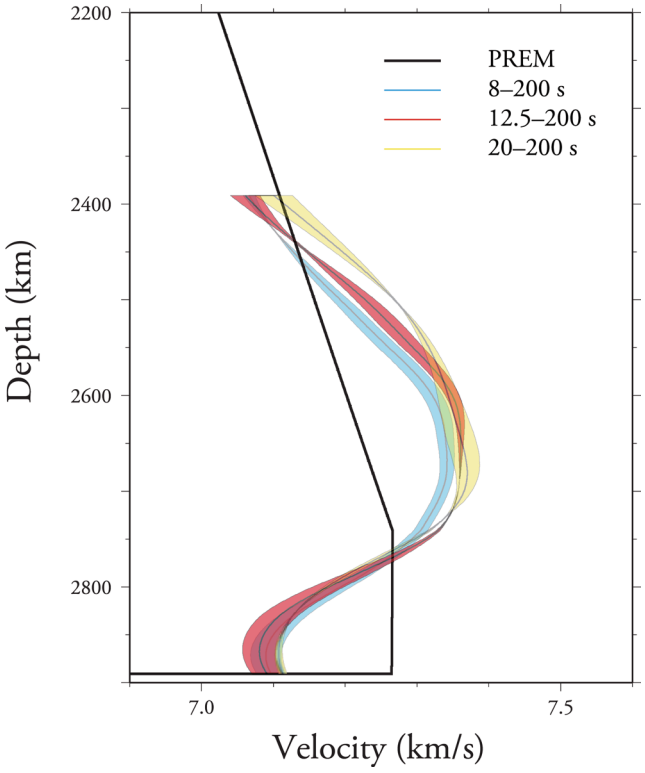


Figure 9. The averages of the models obtained for each of the three passbands (blue: 8–200 s, red: 12.5–200 s and yellow: 20–200 s) by the direct inversion and hybrid method are shown with their $1-\sigma$ standard deviation (obtained by treating the two models as nominally independent). The black curve shows PREM.

and we use the effective number of independent data, n', in place of n in computing AIC.

If we knew the true value of α (i.e. the true amount of redundancy of the data set), using AIC to terminate the inversion would be straightforward. However, unfortunately, we do not know the true value of α, so we show models for various values of α to study how many CG vectors we should use.

2.5 Checkerboard tests

We conduct synthetic resolution ('checkerboard') tests to see if it is possible to resolve various synthetic structure models for the data set of stations and events considered in the actual inversions. We conducted tests for two model perturbations (Fig. 5). Synthetic seismograms were calculated exactly for the two models shown by the black curves and then used as the input 'data' for inversions relative to PREM. The green lines give the best AIC value. However, the models obtained using only two CG vectors appear more reasonable. This may be because AIC is not rigorously applicable to noise-free synthetic data. Although it is difficult to detect a sharp velocity jump because the range of epicentral distances in the data set is relatively long and the passbands are also relatively long-period, we can see that the data set in this study has enough resolution to investigate the target region. Note that our methods invert for the smoothest and simplest model that can explain the observed data.

3 RESULTS

3.1 Comparison of inversion results for different static correction methods

We conduct waveform inversion for the passbands 8–200, 12.5–200 and 20–200 s using the three types of static correction discussed earlier. We show the AIC values and the variance in Fig. 6 and Table 2. The AIC curves and the variance curves for the direct inversion results are different from those for the other inversions. This is quite natural because this method uses more parameters than the others. Although we have to use a larger value of k than for the other two methods, we can see that this method reduces the variance for each data set. In the other two methods, we have similar features of the AIC curves and the variance curves.

Fig. 7 shows the models obtained by our inversions. The model that minimizes AIC is shown in red in each panel. When one additional basis vector is used the models in the panels in the bottom two rows (direct inversion and hybrid) do not change markedly, but the models in the top row (autopick), especially in the top right panel (20–200 s passband), are less stable.

Fig. 8 shows the models with the best AICs for each of the various inversions. Although the depth ranges of the velocity increase (2400–2700 km) and decrease (2700–2900 km) are the same, the models for the autopick method (the top row of Fig. 7) differ significantly from the others. On the other hand, we obtain robust models using the direct inversion method and the hybrid method. The common characteristics of the latter two methods is that the time-shift for data correction is separated into source and receiver

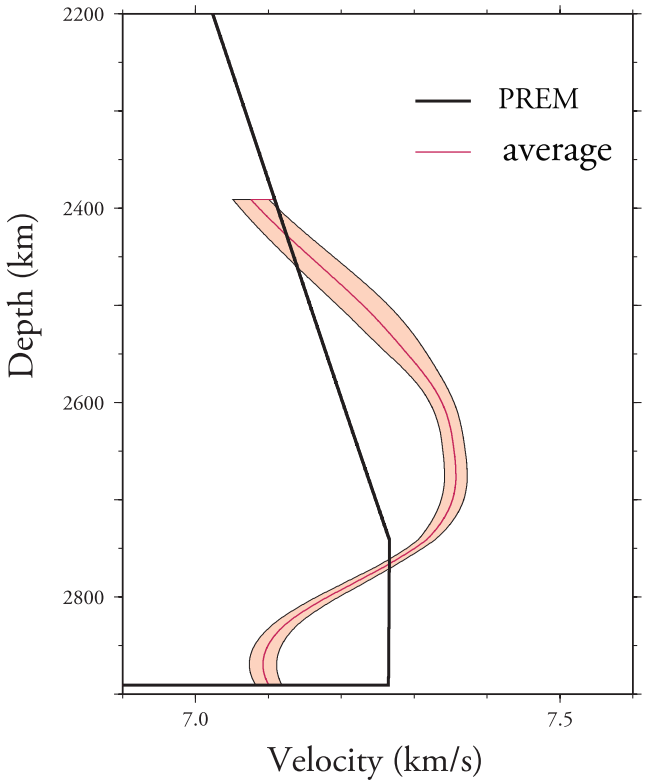


Figure 10. The average of the models for both the direct inversion and the hybrid method are shown by the red curve with the $1-\sigma$ standard deviation shown by the orange region (treating the various models as nominally independent). The black curve shows PREM.

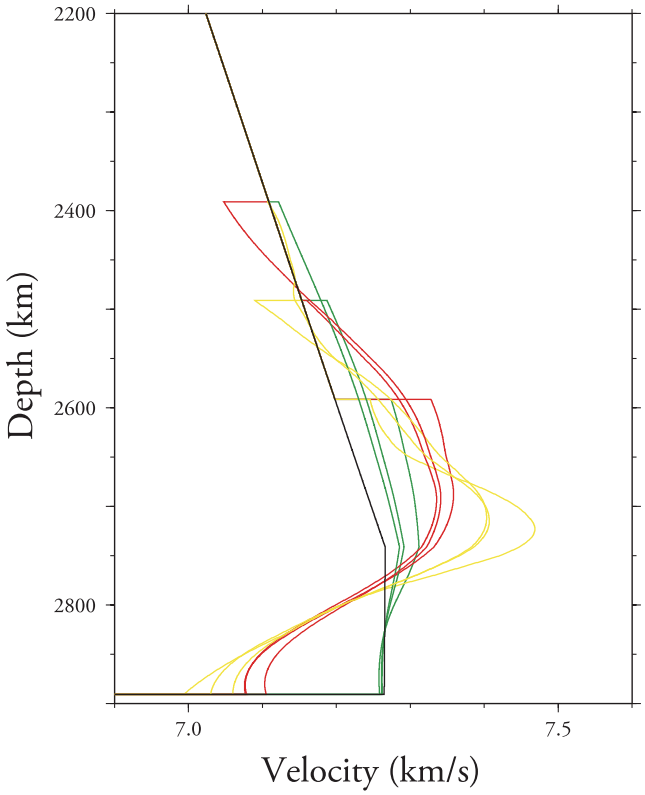


Figure 11. The models for the three different tie-in depths obtained using the hybrid method. Green curves show the models obtained using one CG vector. Red curves show the models obtained using two CG vectors. Yellow curves show the models obtained using three CG vectors.

terms, completely for the direct inversion method and for $\Delta > 80^\circ$ for the hybrid method.

Hereafter, we exclude the models obtained by the autopick method. We consider the models obtained by the other two methods for each of the three passbands of the data set and calculate the average velocity for each method (Fig. 9). Because our data selection excludes data with a poor S/N ratio, the uncertainty due to noise is much smaller than that for the time-shift correction. To see the respective uncertainties, we show the standard deviation for these models. We can see that the deviation is smaller than the perturbations to PREM. We can see no frequency dependence in these velocity models. All of the models have a velocity increase in the depth range between 2400–2600 km and a velocity decrease in the depth range between 2700–2800 km. Fig. 10 shows the average of all six models. The standard deviation does not vary greatly with depth.

We call the depth above which the model is fixed to PREM the 'tie-in depth'. We now check the dependence of the models obtained by the inversion on the choice of tie-in depth. As shown in Fig. 11, we can see that there are no major differences between the models for the various tie-in depths. This suggests that the partial derivatives for each parameter for each depth are sufficiently independent to ensure robustness.

To confirm the reasonableness of the models, we present 'quality control stacks' in Fig. 12. Note that these are merely ancillary visual confirmations of the quantitative results obtained by the inversions. See Kawai *et al.* (2009b) and Konishi *et al.* (2009) for details. We can see an improvement in variance reduction for the time-shifts obtained by inversion using the hybrid method and a further improvement for the model obtained by inversion using the hybrid method. We can see improvement in the earlier part of the *S* waves, whereas the later part, which consists of scattered coda-type phases that cannot be explained by any 1-D model, does not show significant change. We also compare all of the waveforms used in Fig. 13. In the epicentral distance range 75°–90°, we can see that the shape of the *S* phase for the obtained model for the 8–200 s passband is better than that for PREM.

3.2 Possibility of laterally heterogeneous structure

In this study, as shown in Fig. 2, the ray paths cover a large area under central Asia. As shown in Figs 2 and 14, we divide the whole region into two subregions to see possible lateral heterogeneity. The obtained models are broadly similar, but the deepest parts have a slight difference. We also compared our models to SYL1 (Young & Lay 1987) and SGLE (Gaherty & Lay 1992), respectively. Although the covered regions do not match exactly, they overlap substantially. However, as shown by Fig. 14b, our model and SGLE have large differences at depths greater than 2800 km. According to Wysession *et al.* (1992), the lowermost 190 km of the mantle beneath Indonesia has higher velocity than PREM. Thus there may be lateral differences not only across but also along the ray paths used in this study. In the future, it would be desirable to conduct waveform inversion for the 3-D structure of this region, but, as the number of usable seismograms is small, this may be difficult.

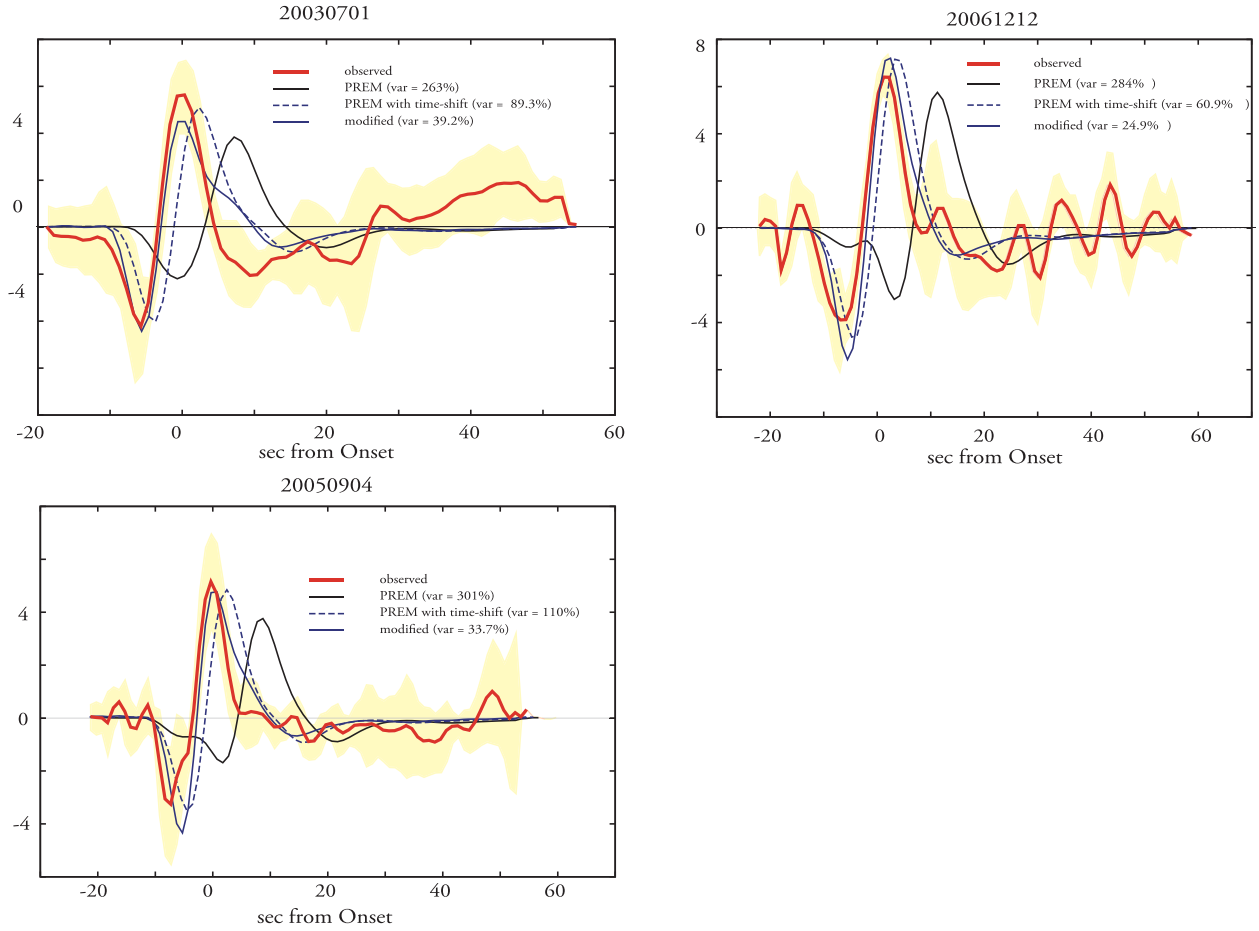


Figure 12. Quality control (QC) stacks of the waveforms for each of the three events. Note that these are merely ancillary visual confirmation of the reasonableness of the inversion results. (See Kawai *et al.* 2009b; Konishi *et al.* 2009, for details.) The origin time is the *S*-arrival time (after making static corrections to the data). The waveforms are stacked with normalized maximum amplitude (the unit of vertical axis is in $\mathrm{m\,s^{-1}}$). The time windows of the stacked waveforms are those used in the inversion. The later part of the time window consists of scattered coda waves for which no model can be expected to fit the data. The red curves show the stacked observed waveforms with the 1σ standard deviation shown by the yellow region (treating all the points as nominally independent) and the black curves show the stacked synthetic waveforms for PREM. The dotted curve is computed for the time-shifts obtained using the hybrid method. Dark blue curves are stacked synthetic waveforms for the model obtained by the inversion for the 8–200 s passband data set using the hybrid method. All of the waveforms are filtered in the passband 8–200 s. The variance of the residuals are shown for each stacked waveform.

4 DISCUSSION

We conducted waveform inversion for the radial shear velocity structure of the lowermost mantle beneath the Indian subcontinent and Tibetan Plateau. This is the first body wave waveform inversion study for a region in the lowermost mantle whose average velocity is close to that of PREM (e.g. Takeuchi 2007).

Our study considered three different methods for making static corrections for the effect of structure outside the target region and showed robust results for the hybrid method, which we developed in this study, and the direct inversion method. To observe frequency dependence, we conducted inversions with data sets for three passbands and found that the obtained models are very similar. We confirmed the improvement of the fit of the synthetic and observed waveforms by presenting quality control stacks (Fig. 12). We also confirmed that the tie-in depth is not a critical parameter for the inversion in this study (Fig. 8).

Thomas *et al.* (2004) suggested that there is a sharp velocity increase in the lowermost several hundred kilometres of the mantle, with a sharp velocity reduction approximately 80 km above the CMB. Bilek & Lay (1998) also examined lower mantle structure beneath central Eurasia. They suggest two scatterers, as was reported by Gaherty & Lay (1992). The suggested possible scatterer locations are near the north edge of our study region. The velocity increase and decrease in our model may correspond to those anomalies. On the other hand, Weber & Davis (1990) also suggested a model with a positive velocity jump at a depth of around 2600 km, but without a negative velocity jump at greater depths. However, they also noted that the *ScS* traveltime for the SWDK model is several seconds earlier than the observed arrivals. Although they did not mention this possibility, a velocity decrease below the inferred velocity jumps for *SdS* could explain the observed delay of *ScS* without requiring changes in the suggested velocity jumps for *SdS*.

As discussed earlier in Section 2.4, our methods cannot directly invert for a sharp velocity discontinuity. If the velocity increase and decrease in our model were converted to the sharp velocity discontinuity parametrization used by other studies, our model would correspond to a positive velocity discontinuity 200–300 km above the CMB and a negative discontinuity 100–150 km above the CMB. Although, as shown in Fig. 7 and discussed in Section 3, we also see velocity increases in models with three CG vectors, those models do not give the best AIC.

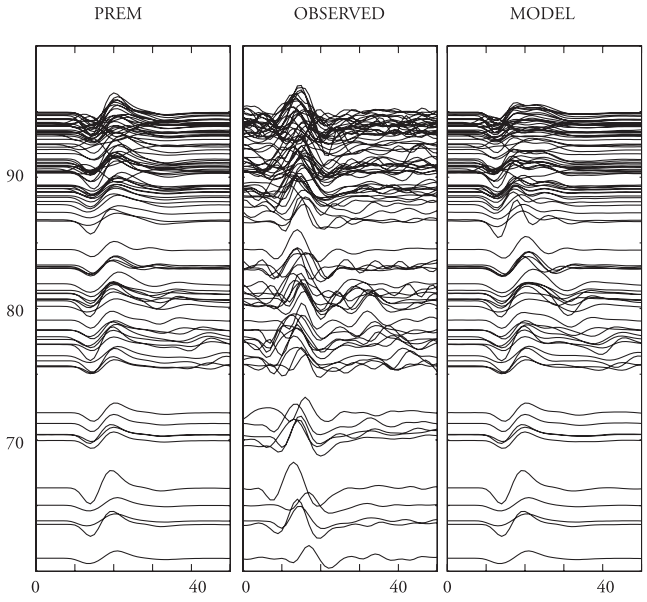


Figure 13. We align all of the waveforms used for all of the events. Each time window is the one that we used in this study (discussed in Section 2.1.1). The vertical axis is epicentral distance. (Left-hand panel) Synthetic waveforms for the initial model PREM. (Central panel) Observed seismograms. (Right-hand panel) Modified waveforms.

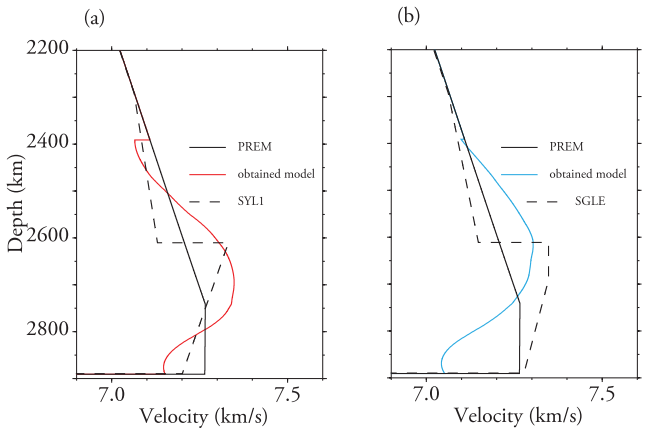


Figure 14. Results for the two subregions. The sampled regions are shown by the red and sky blue ray paths in Fig. 2. In the left-hand and right-hand panels, we also show models SYL1 (Young & Lay 1987) and SGLE (Gaherty & Lay 1992), respectively, for purposes of comparison.

The average composition in the mantle is thought to contain impurities such as aluminum and iron (Ringwood 1962). Arguments from mineral physics show that impurities in Mg–pv decrease the shear wave velocity (Tsuchiya & Tsuchiya 2006). For example, a 1 mol per cent increase in the amount of both aluminum and iron causes a 0.303 per cent velocity decrease in Mg–pv or a 0.367 per cent velocity decrease in Mg–ppv. In this study we estimated the shear wave velocity to be about 7.11 km s^{-1} immediately above the CMB. As the temperature at the CMB is isothermal, the shear wave velocity at the CMB would be equal everywhere if the mineralogical phase and chemical composition were identical (Kawai & Tsuchiya 2009). Since waveform inversion is capable of inferring seismic velocity directly, we can distinguish the effects of temperature from those of chemical composition.

The error bars in Fig. 10 show the nominal uncertainty of the models obtained by our inversions. However, we can see the robustness between the results obtained by several methods. It thus seems reasonable to assume that the models have quantitative reliability, especially immediately above the CMB where there is good ray path (*ScS* or *S*) coverage. Our previous study of D″ beneath central America (Kawai *et al.* 2007a) found an *S*-wave velocity of about 7.25 km s^{-1} immediately above the CMB, which is about 0.14 km s^{-1} faster than the velocity obtained in this study. The difference between the values found here and those of Kawai *et al.* (2007a) seem significant. Although the size of the data set and the methodologies differ somewhat between the two studies, as Kawai *et al.* (2007a) also used *S* and *ScS* waves, we can compare the velocities immediately above the CMB. The observed discrepancy could be explained by either chemical differences or thermal differences, but if we assume there are no differences in chemical and thermal conditions, it can be interpreted as due to the amount of impurities in Mg-ppv of the study region being 6 mol per cent larger than that beneath central America, on the assumption that the ratio of aluminum and iron is constant in Mg-ppv. As PREM can be approximated as pyrolitic composition with 7 mol per cent impurities (McDonough & Sun 1995), we therefore estimate the amount of impurities in Mg–ppv beneath the Indian subcontinent and Tibetan Plateau to be 13 mol per cent. Impurities in D″ would cause a two-phase coexistence region for the pv to ppv phase transition (e.g. Tsuchiya & Tsuchiya 2008). This might cause a broad gradient velocity increase in the depth range from 2400–2600 km. As some studies have also reported *P*-wave reflections from this depth range, the possible existence of sharp reflectors cannot be excluded.

In conclusion, we obtained robust models for the lowermost mantle beneath the Indian subcontinent and Tibetan Plateau having a velocity increase in the depth range 2400–2600 km and a decrease in the depth range 2700–2800 km. This suggests that the average velocity in the depth range 2391–2891 km (i.e. in the lowermost 500 km of the mantle) is about 7.2 km s^{-1} and that the pv to ppv phase transition occurs at a depth of about 2400–2600 km. According to Kawai & Tsuchiya (2009), this model can be explained by a 200–300 km thick thermal boundary layer and a 3800 K core temperature for the velocity profile for pyrolite.

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